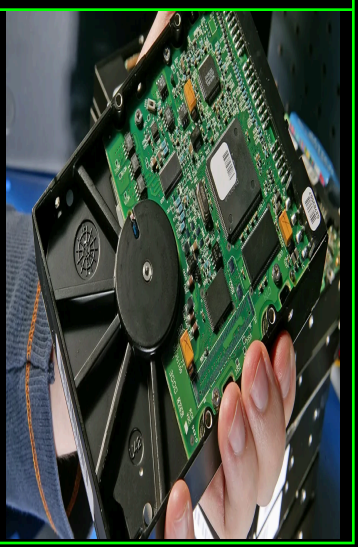
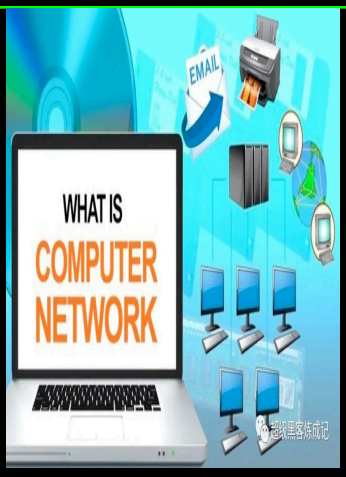
Introduction to Computers

Computer Terms :

	Defnition	Examples	Image
Hardware	Hardware refers to the physical components of a computer system that you can see and touch. This includes the central processing unit (CPU), memory modules, hard drives, keyboards, mice, monitors, and other peripherals. Hardware plays a key role in enabling the functionality and operation of a computer system.	CPU RAM Storage devices (hard drives, solid-state drives), motherboards, graphics cards, power supplies	
Software	Programming and controlling hardware through software to achieve	Operating system (Windows, macOS), driver	

	virtualization, flexible customization and automated management, thereby improving system efficiency and providing customized services		
Input Devices	Any hardware device that sends data to a computer and allows the user to interact with and control it	Keyboard, mouse, microphone, camera	
Output Devices	A type of computer hardware that converts computer-processed data into a format that can be understood by the human eye, ear, or other senses for display, printing, playing sound, or controlling other devices.	Monitors, printers, speakers (audio), projectors	

Peripheral Devices	Hardware components that connect to a computer to expand its functionality, but are not core to its main operation.	Input devices (such as keyboard, mouse, scanner), output devices (such as monitor, printer, speakers), storage devices (such as external hard disk, USB flash drive), network devices (such as router, network card), and other auxiliary devices (such as webcam, drawing tablet, joystick).	
Memory	Computer memory is the hardware that a computer uses to store data and instructions for quick access by the processor.	Primary Memory (RAM) Secondary Memory (Storage)	

Networking	The practice of connecting two or more computing devices to allow them to communicate, exchange data, and share resources like files and printers.	**Personal Area Network (PAN)** **Local Area Network (LAN):**	
WiFi	Wifi is a family of wireless network protocols based on the IEEE &02.11 family of standards.	**Wireless Local Area Network (WLAN)**	

Data Storage Devices :

	Description	Speed and Capacity	Image

Hard Drive	Uses spinning magnetic disks to store data, cheaper per GB.	Slower speed (80–160 MB/s), large capacity (up to 20TB).	
Solid State Drive (SSD)	Stores data on flash memory, no moving parts, more durable.	Very fast (500 MB/s – 7000 MB/s), medium to large capacity (256GB–8TB).	
Hybrid SSD	Combines HDD with a small SSD cache to improve performance.	Faster than HDD but slower than SSD, 500GB–2TB.	
CD	Optical disc used mainly for music/software.	Low speed, capacity ~700MB.	

DVD	Optical disc used for movies and data storage.	Capacity 4.7GB (single layer), 8.5GB (dual layer).	
Blu-ray	High-defnition optical disc, mainly for movies/games	Capacity 25GB (single), 50GB (dual).	
Flash Drive Memory	Portable USB device using flash memory.	Speed varies (30MB/s – 400MB/s), capacity 2GB–2TB.	

Communication Devices :

Type	Characteristics	Image
Modem	Converts digital data to analog signals for internet over telephone/cable lines.	